A3-PS003-TC222

FedCE: Communication-Efficient Wireless Traffic Prediction with Federated Learning

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Problem Description

Centralized traffic forecasting does not scale, and federated learning still leaves two gaps.

- Predicting wireless traffic lets operators provision capacity ahead of congestion and put lightly used stations into energy-saving modes.
- Pooling every station's data on one server raises four problems: privacy of raw records, bandwidth cost, limits on data sharing, and load on a single server.
- Federated learning keeps raw data on site, but local traffic data are rarely alike (client drift) and every training round costs communication.
- FedCE targets both. On the Trento dataset it cuts test MSE from 4.37 (FedAvg) to 1.56 while sending 10% of the gradient information.

Background

How federated learning and the traffic model work.

- The server sends a shared model to each station; each station trains on its own data; only model updates come back. Raw records never leave the station.
- This limits data movement, but model updates alone do not guarantee privacy.
- System model: K base stations and one coordinating server. Each station keeps its own traffic sequence.
- A sliding window turns recent history into model inputs; the next traffic value is the prediction target.
- Closeness captures recent values (traffic at 4:00 p.m. depends on 3:50 and 3:40).
- Period captures repetition (Friday 4:00 p.m. resembles Thursday 4:00 p.m.).

Literature & RFC Review

One primary paper and five supporting works; baselines are Standalone, Centralized, FedAvg and FedDA.

Reference	Contribution
Gao et al., 2024 [1]	Primary paper: FedCE, communication-efficient federated traffic prediction
Konečný et al., 2016 [2]	Federated optimization for on-device intelligence
Li et al., 2020 [3]	Federated learning: challenges, methods, future directions
Karimireddy et al., 2020 [4]	SCAFFOLD, the basis of FedCE's gradient correction
Lin et al., 2017 [5]	Deep gradient compression to cut bandwidth
Kingma & Ba, 2014 [6]	Adam, the optimizer used in the experiments

Dataset: Milano and Trento call detail records (Harvard Dataverse). Access and reuse follow the dataset's stated terms.

Proposed Solution

Each round: share the model, train locally, correct and compress, then aggregate by similarity.

- Gradient correction: control variables on the server and each client nudge every local step toward the global objective, limiting client drift (builds on SCAFFOLD [4]).
- Top-K compression: each station sends only the K largest gradient entries; unsent values are retained so they are not lost for good.
- Adaptive aggregation: updates are weighted by Pearson correlation of client gradients combined with the reciprocal of station distance, so similar, nearby clients carry more influence.
- In the paper's example, clients form two correlated groups (within-group 0.78 to 0.96 and 0.56 to 0.87), one outlier (at most 0.38), and at most 0.44 across groups.
- Communication percentages (10%, 20%, 40%) count only gradient information sent during training, relative to federated baselines. They are not total network traffic or energy.

Solution Optimization: Evaluation Setup

Call detail records, 1 Nov 2013 to 1 Jan 2014; each coverage cell is one federated client.

	Milano	Trento
Cells	10,000	6,575
Layout	100 × 100 grid	Hourly series per cell
Original interval	10 minutes	Resampled to 1 hour

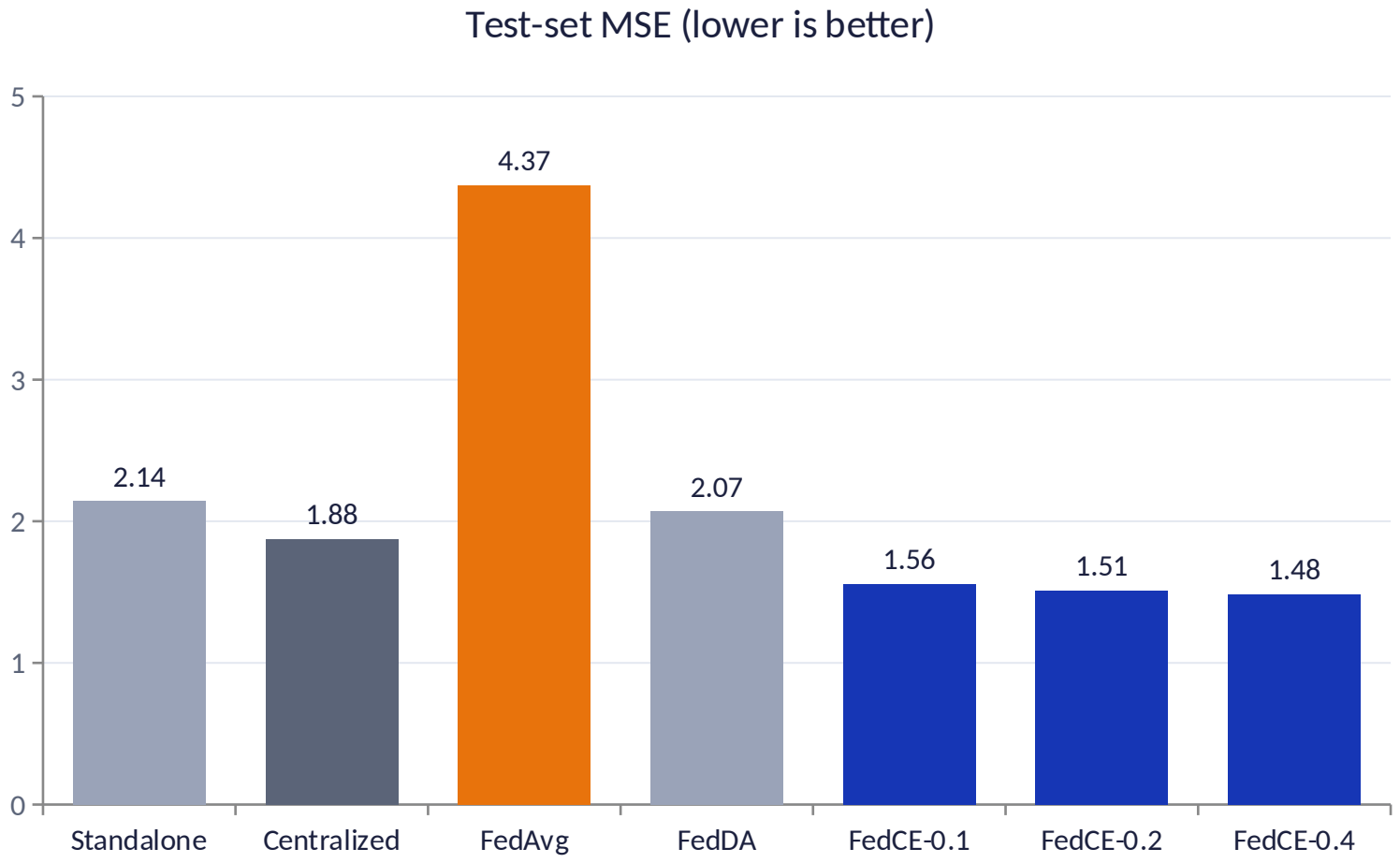
- 100 randomly selected base stations; time-ordered split of 7/8 for training and 1/8 for testing.
- Lightweight MLP, Adam optimizer, learning rate 0.001, batch size 20, 1 local epoch, 100 global rounds.
- Compared: Standalone, Centralized, FedAvg, FedDA, and FedCE at 0.1, 0.2 and 0.4.
- Caution: results from 100 stations may not carry over to a full deployment.

Optimization Proposed: Before and After

Trento test set	MSE ↓	MAE ↓	R^2 ↑	Communication
Standalone	2.1411	0.7751	0.7631	n/a
Centralized	1.8761	0.7300	0.8550	n/a
FedAvg (before)	**4.3719**	**1.1072**	**0.6621**	**100%**
FedDA	2.0716	0.7632	0.8399	100%
FedCE-0.1 (after)	**1.5590**	**0.7164**	**0.8795**	**10%**
FedCE-0.2	1.5108	0.6968	0.8832	20%
FedCE-0.4	1.4844	0.6867	0.8853	40%

- FedAvg to FedCE-0.1: MSE 4.3719 to 1.5590 (−64%), MAE 1.1072 to 0.7164 (−35%), R^2 0.6621 to 0.8795 (+0.22), communication 100% to 10% (−90%). Percentages are derived from the table.
- FedCE-0.4 has the lowest errors and highest R^2 at 40% communication; FedCE-0.1 trades a little accuracy for 10%.

Results: Test MSE on Trento



- 10% of baseline communication already beats centralized MSE (1.56 vs 1.88).
- Going from 10% to 40% communication buys only about 5% lower MSE (1.559 to 1.484).
- FedDA, the best federated baseline (2.07), trails every FedCE variant.

Milano Results and Limitations

Milano: centralized training holds the strongest listed metrics; FedCE's accuracy shifts with compression and its main edge is lower communication.

Strengths

- Raw traffic records stay at each station.
- Top-K compression cuts the gradient values transmitted.
- Correction targets client drift; aggregation uses similarity and distance.
- Evaluated against several methods on two real datasets.

Limitations

- Privacy is improved, not guaranteed; secure aggregation or differential privacy may be needed.
- Stronger compression can hurt accuracy or slow convergence.
- Only 100 randomly chosen stations were tested, not a full network.
- Holidays, outages and events can shift traffic patterns.

Solution Architecture & Design

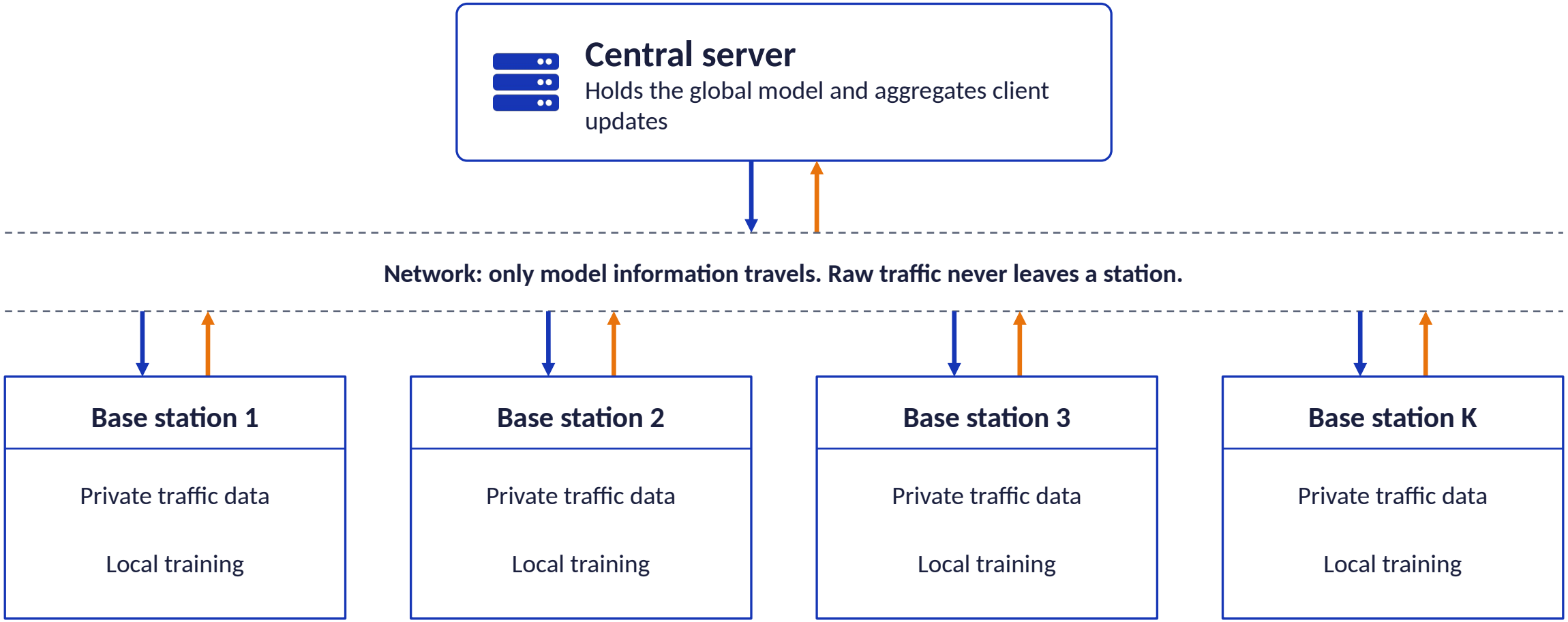


Figure: System Architecture UML Diagram. Blue arrows: global model and control information down; orange arrows: compressed updates up.

Data Flow Diagram

1. Broadcast
Server

Sends the global model and control information.

Data: Global model, control info

2. Train
Stations

Each station trains on its own traffic data.

Data: Local update

3. Compress
Stations

Correct the gradient and keep the Top-K entries.

Data: Sparse update, residual kept

4. Aggregate
Server

Weighs gradient similarity and station distance.

Data: Similarity-weighted sum

5. Update
Server

Builds the new global model.

Data: New global model

Repeat every round until the stopping condition is met

Compression targets communication volume; correction targets client drift; adaptive aggregation targets unequal station relationships.

Deployment Timeline

Four phases, changing one thing at a time and comparing against the previous baseline.

Phase	Activities	Milestone
1. Baseline	Implement FedAvg with a small number of clients; record bytes exchanged each round.	Baseline bytes per round logged
2. Increments	Add one improvement at a time: correction, Top-K, adaptive aggregation.	Gain attributed to each change
3. Hardening	More cities and periods (weekends, holidays, events); client dropout, slow devices, missing data; secure aggregation or differential privacy.	Robustness and privacy cost measured
4. Comparison	Top-K vs quantization, threshold and error-feedback schemes; MLP vs LSTM and temporal convolutional models; account for bytes, latency, compute, energy.	Full cost model

Conclusion

FedCE sends 10% to 40% of the gradient information and leads on Trento; Milano is more conditional.

- Trento: all three FedCE variants beat every listed method on MSE, MAE and R^2.
- Milano: the trade-off is less favourable; quality depends on the dataset and the compression setting.
- Weigh communication efficiency, accuracy and privacy together. Federated learning reduces the need to pool raw data, but performance and privacy still need deployment-specific evaluation.

Bibliography

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Thank You for Your Attention!